EI331 Signals and Systems

Lecture 28

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Contents

DT System Function

Recall the response of a DT LTI system to the input $x[n]$ is

$$y[n] = (x * h)[n]$$

where h is the impulse response of the system.

If x and h have z-transforms, the convolution property implies

$$Y(z) = X(z)H(z)$$

in their common ROC.

If the ROC has a nonempty interior point, the system function (aka transfer function) $H(z)$ uniquely determines h and hence system properties through the Laurent series expansion

$$H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n}$$

Causality

Recall h is right-sided iff its ROC is the exterior of a circle, i.e. $R_1 < |z| < \infty$

$$H(z) = \sum_{n=N_1}^{\infty} h[n]z^{-n}$$

The following conditions are equivalent

1. $N_1 \geq 0$
2. $H(z^{-1}) = \sum_{n=N_1}^{\infty} h[n]z^n$ is a convergent power series on $|z| < \frac{1}{R^1}$
3. 0 is a removable singularity of $H(z^{-1})$
4. ∞ is removable singularity of $H(z)$
5. $\lim_{z\to 0} H(z^{-1})$ exists and is finite
6. $\lim_{z\to\infty} H(z)$ exists and is finite[1]

[1] This is the more precise statement of $\infty \in$ ROC.

Causality

An LTI system with system function $H(z)$ is causal iff

1. the ROC is the exterior of a circle
2. $\lim_{z \to \infty} H(z)$ exists and is finite

$$\boxed{\textbf{causal} \iff \textbf{ROC is the exterior of a circle including } \infty}$$

An LTI system with rational system function $H(z) = \frac{N(z)}{D(z)}$ is causal iff

1. the ROC is $|z| > |p|$, where p is the outermost pole
2. $\deg D \geq \deg N$

Example. $H(z) = \frac{z^3 - 2z^2 - z}{z^2 + \frac{1}{4}z + \frac{1}{8}}$ cannot be the system function of a causal system.

Stability

Recall an LTI system is stable iff its impulse response $h \in \ell_1$, i.e.

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

i.e. $H(z)$ converges absolutely on the unit circle $|z| = 1$, so its ROC $R_1 < |z| < R_2$ must satisfy $R_1 < 1 < R_2$.

$$\boxed{\textbf{stable} \iff \textbf{ROC includes the unit circle } |z| = 1}$$

A **causal** LTI system with rational system function $H(z)$ is stable iff all its poles are inside the unit circle.

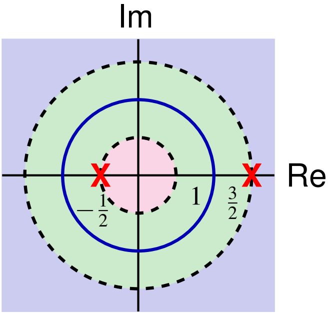
Example. A causal system with $H(z) = \frac{1}{1-az^{-1}}$ is stable iff $|a| < 1$

Example. A system with $H(z) = \frac{1}{1-az^{-1}}$ where $|a| > 1$ and ROC $|z| < |a|$ is also stable, but it is noncausal.

Example

$$H(z) = \frac{1}{2}\left[\frac{z}{z-\frac{3}{2}} - \frac{z}{z+\frac{1}{2}}\right]$$

There are two poles $p_1 = -\frac{1}{2}$ and $p_2 = \frac{3}{2}$.



1. $|z| > \frac{3}{2}$, causal, unstable

$$h_1[n] = \frac{1}{2}\left(\frac{3}{2}\right)^n u[n] - \frac{1}{2}\left(-\frac{1}{2}\right)^n u[n]$$

2. $\frac{1}{2} < |z| < \frac{3}{2}$, noncausal, stable

$$h_2[n] = -\frac{1}{2}\left(\frac{3}{2}\right)^n u[-n-1] - \frac{1}{2}\left(-\frac{1}{2}\right)^n u[n]$$

3. $|z| < \frac{1}{2}$, noncausal, unstable

$$h_3[n] = -\frac{1}{2}\left(\frac{3}{2}\right)^n u[-n-1] + \frac{1}{2}\left(-\frac{1}{2}\right)^n u[-n-1]$$

Linear Constant-coefficient Difference Equations

LTI system with input and output related by

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

Take z-transform of both sides

$$\sum_{k=0}^{N} a_k z^{-k} Y(z) = \sum_{k=0}^{M} b_k z^{-k} X(z)$$

so

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

System function is always rational

Difference equation does not specify ROC! Need additional conditions (e.g. stability, causality) to determine $h[n]$.

Example

Consider LTI system with input and output related by

$$y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{1}{3}x[n-1]$$

System function

$$H(z) = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

Two possibilities for ROC: $|z| > \frac{1}{2}$ and $|z| < \frac{1}{2}$.

1. If $|z| > \frac{1}{2}$

$$\frac{1}{1 - \frac{1}{2}z^{-1}} = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} \xleftrightarrow{z^{-1}} \left(\frac{1}{2}\right)^n u[n]$$

By linearity and time-shifting property,

$$h[n] = \left(\frac{1}{2}\right)^n u[n] + \frac{1}{3}\left(\frac{1}{2}\right)^{n-1} u[n-1]$$

Example (cont'd)

Consider LTI system with input and output related by

$$y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{1}{3}x[n-1]$$

System function

$$H(z) = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

Two possibilities for ROC: $|z| > \frac{1}{2}$ and $|z| < \frac{1}{2}$.

2. If $|z| < \frac{1}{2}$

$$\frac{1}{1 - \frac{1}{2}z^{-1}} = \frac{-2z}{1 - 2z} = -\sum_{n=1}^{\infty} 2^n z^n \overset{z^{-1}}{\longleftrightarrow} -\left(\frac{1}{2}\right)^n u[-n-1]$$

By linearity and time-shifting property,

$$h[n] = -\left(\frac{1}{2}\right)^n u[-n-1] - \frac{1}{3}\left(\frac{1}{2}\right)^{n-1} u[-n]$$

Example (cont'd)

Consider LTI system with input and output related by

$$y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{1}{3}x[n-1]$$

If we use Fourier transform, then frequency response is

$$H(e^{j\omega}) = \frac{1 + \frac{1}{3}e^{-j\omega}}{1 - \frac{1}{2}e^{-j\omega}}$$

and

$$h[n] = \left(\frac{1}{2}\right)^{n} u[n] + \frac{1}{3}\left(\frac{1}{2}\right)^{n-1} u[n-1]$$

Why only one possibility?

- Fourier transform method assumes stability, requiring that ROC of $H(z)$ contain the unit circle, i.e. $|z| > \frac{1}{2}$
- In general, not applicable to unstable systems

Example

Consider LTI system with input and output related by

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]$$

System function

$$H(z) = \frac{2}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}$$

By partial fraction expansion

$$H(z) = \frac{2}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})} = \frac{4}{1 - \frac{1}{2}z^{-1}} - \frac{2}{1 - \frac{1}{4}z^{-1}}$$

Take inverse z-transform to obtain impulse response

$$\begin{cases} h_1[n] = 4\left(\frac{1}{2}\right)^n u[n] - 2\left(\frac{1}{4}\right)^n u[n], & \text{ROC: } |z| > \frac{1}{2} \\ h_2[n] = -4\left(\frac{1}{2}\right)^n u[-n-1] - 2\left(\frac{1}{4}\right)^n u[n], & \text{ROC: } \frac{1}{4} < |z| < \frac{1}{2} \\ h_3[n] = -4\left(\frac{1}{2}\right)^n u[-n-1] + 2\left(\frac{1}{4}\right)^n u[-n-1], & \text{ROC: } |z| < \frac{1}{4} \end{cases}$$

Example (cont'd)

Find response to $x[n] = \left(\frac{1}{4}\right)^n u[n]$.

$$X(z) = \frac{1}{1 - \frac{1}{4}z^{-1}}, \quad |z| > \frac{1}{4}$$

z-transform of response

$$Y(z) = H(z)X(z) = \frac{2}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})} \cdot \frac{1}{1 - \frac{1}{4}z^{-1}},$$

NB. Output well-defined only for $|z| > \frac{1}{2}$ and $\frac{1}{4} < |z| < \frac{1}{2}$!

Exercise. Verify that $h_3 * x$ is not well-defined.

By partial fraction expansion (see slides 20-22 of Lecture 19)

$$Y(z) = -\frac{4}{1 - \frac{1}{4}z^{-1}} - \frac{2}{(1 - \frac{1}{4}z^{-1})^2} + \frac{8}{1 - \frac{1}{2}z^{-1}}$$

Take inverse z-transform to obtain response for first two cases.

Inverse Z-transform of $\frac{1}{(1-az^{-1})^n}$

Recall

$$\frac{1}{1-a\zeta} = \sum_{n=0}^{\infty} a^n \zeta^n, \quad |a\zeta| < 1$$

and

$$\frac{1}{1-a\zeta} = -\sum_{n=-\infty}^{-1} a^n \zeta^n, \quad |a\zeta| > 1$$

Repeat the derivation on slide 23 of Lecture 19, and let $\zeta = z^{-1}$,

$$\binom{n+m-1}{m-1} a^n u[n] \xleftrightarrow{z} \frac{1}{(1-az^{-1})^m}, \quad |z| > |a|$$

and

$$(-1)^m \binom{-n-1}{m-1} a^n u[-n-m] \xleftrightarrow{z} \frac{1}{(1-az^{-1})^m}, \quad |z| < |a|$$

Example

Causal LTI system described by

$$y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{2}{3}x[n-1]$$

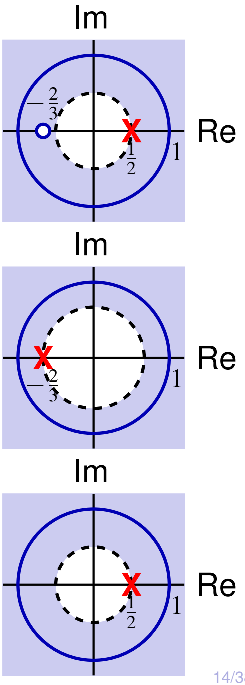
Find response to $x[n] = (-\frac{2}{3})^n u[n]$

$$H(z) = \frac{1 + \frac{2}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}$$

$$X(z) = \frac{1}{1 + \frac{2}{3}z^{-1}}, \quad |z| > \frac{2}{3}$$

$$Y(z) = \frac{1}{1 - \frac{1}{2}z^{-1}},\ |z| > \frac{1}{2} \implies y[n] = \left(\frac{1}{2}\right)^n u[n]$$

ROC of Y is larger than intersection of ROCs of X and H due to pole-zero cancellation at $z = -\frac{2}{3}$.



Example

Suppose an LTI system satisfies the following,

1. output for input $x_1[n] = (\frac{1}{6})^n u[n]$ is $y_1[n] = [a(\frac{1}{2})^n + 10(\frac{1}{3})^n]u[n]$
2. output for input $x_2[n] = (-1)^n$ is $y_2[n] = \frac{7}{4}(-1)^n$

Want to find $H(z)$.

$$X_1(z) = \frac{1}{1 - \frac{1}{6}z^{-1}}, \quad |z| > \frac{1}{6}$$

$$Y_1(z) = \frac{a}{1 - \frac{1}{2}z^{-1}} + \frac{10}{1 - \frac{1}{3}z^{-1}}, \quad |z| > \frac{1}{2}$$

System function

$$H(z) = \frac{Y_1(z)}{X_1(z)} = \frac{[(a+10) - (5 + \frac{a}{3})z^{-1}][1 - \frac{1}{6}z^{-1}]}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})},$$

By 2, $H(-1) = \frac{7}{4} \implies a = -9 \implies H(z) = \dfrac{(1 - 2z^{-1})(1 - \frac{1}{6}z^{-1})}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}$

Example (cont'd)

Three possibilities for ROC of $H(z) = \dfrac{(1-2z^{-1})(1-\frac{1}{6}z^{-1})}{(1-\frac{1}{2}z^{-1})(1-\frac{1}{3}z^{-1})}$

- $|z| < \frac{1}{3}$
- $\frac{1}{3} < |z| < \frac{1}{2}$
- $|z| > \frac{1}{2}$

By the convolution property, ROC of Y_1 contains the intersection of the ROCs of X_1 and $H \implies$ ROC of H is $|z| > \frac{1}{2} \implies$ stable.

Also $H(\infty) = 1$ is finite $\implies$ causal

$$H(z) = \frac{1 - \frac{13}{6}z^{-1} + \frac{1}{3}z^{-2}}{1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}}$$

Difference equation

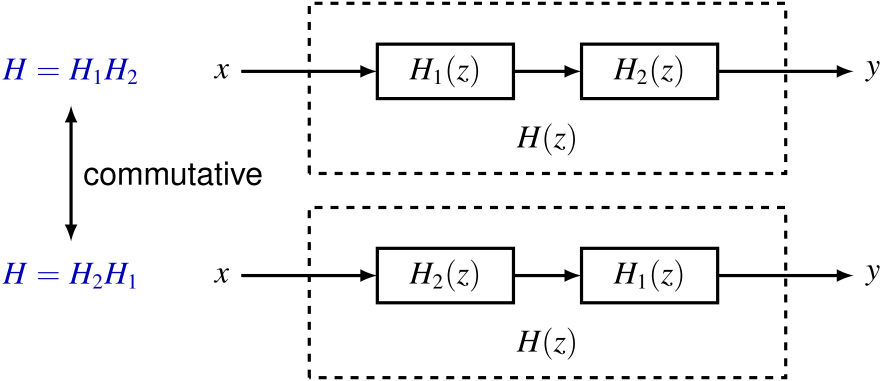
$$y[n] - \frac{5}{6}y[n-1] + \frac{1}{6}y[n-2] = x[n] - \frac{13}{6}x[n-1] + \frac{1}{3}x[n-2]$$

Contents

System Interconnections

LTI systems in series connection

$$Y = (XH_1)H_2 = X(H_1H_2) = (XH_2)H_1$$

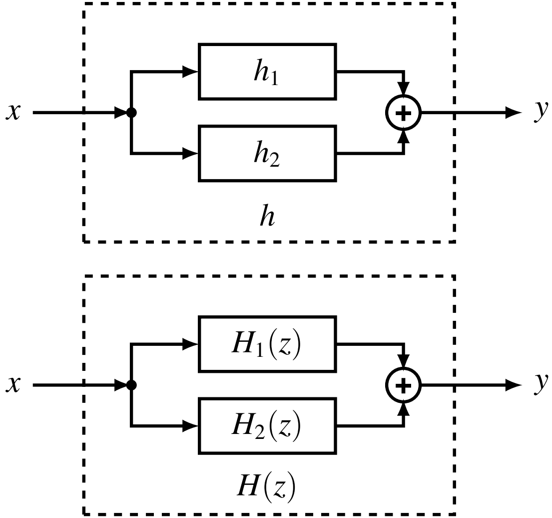


System Interconnections

LTI systems in systems in parallel connection

$h = h_1 + h_2$

$H = H_1 + H_2$

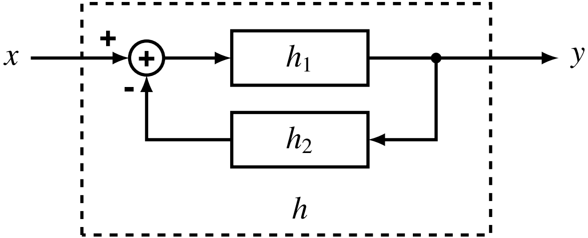


System Interconnections

LTI systems in systems in feedback connection

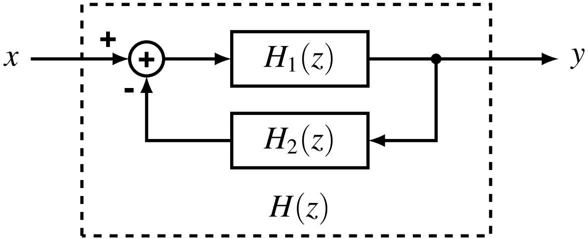
$$y = h_1 * (x - h_2 * y)$$

$$h = ?$$



$$Y = H_1 X - H_1 H_2 Y$$

$$H = \frac{Y}{X} = \frac{H_1}{1 + H_1 H_2}$$



Example

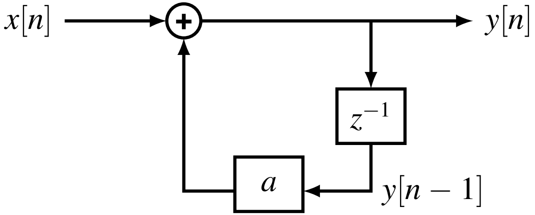
Causal LTI systems with system function

$$H(z) = \frac{1}{1 - az^{-1}}, \quad |z| > |a|$$

Equivalent description by difference equation

$$y[n] - ay[n-1] = x[n]$$

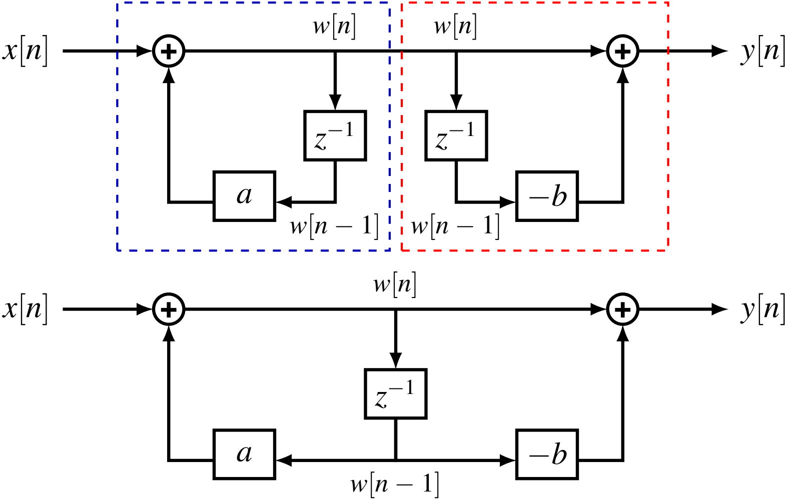
with initial rest condition.



Example

Causal LTI systems with system function

$$H(z) = \frac{1 - bz^{-1}}{1 - az^{-1}} = \left(\frac{1}{1 - az^{-1}}\right)(1 - bz^{-1}), \quad |z| > |a|$$



Example

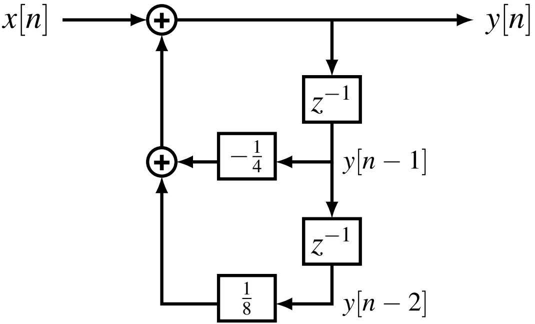
Causal LTI systems with system function

$$H(z) = \frac{1}{(1+\frac{1}{2}z^{-1})(1-\frac{1}{4}z^{-1})} = \frac{1}{1+\frac{1}{4}z^{-1}-\frac{1}{8}z^{-2}}$$

Equivalent description by different equation

$$y[n] + \frac{1}{4}y[n-1] - \frac{1}{8}y[n-2] = x[n]$$

Direct form

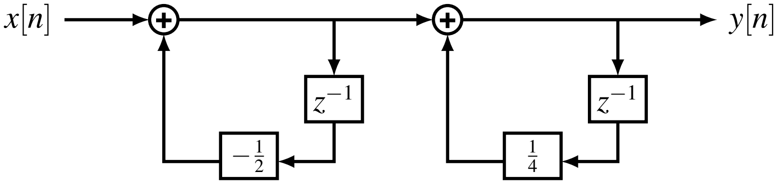


Example (cont'd)

Causal LTI systems with system function

$$H(z) = \frac{1}{1 + \frac{1}{2}z^{-1}} \cdot \frac{1}{1 - \frac{1}{4}z^{-1}}$$

Cascade form

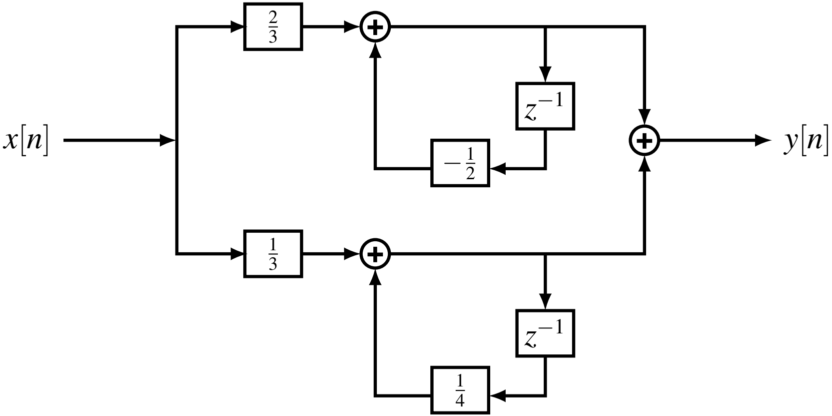


Example (cont'd)

Causal LTI systems with system function

$$H(z) = \frac{2/3}{1 + \frac{1}{2}z^{-1}} + \frac{1/3}{1 - \frac{1}{4}z^{-1}}$$

Parallel form



Contents

Unilateral $\mathcal{Z}$-transform

The (bilateral) z-transform of a DT signal x

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

The unilateral z-transform of x is

$$\mathcal{X}(z) = \sum_{n=0}^{\infty} x[n]x^{-n}$$

also denoted

$$\mathcal{X} = \mathcal{UZ}\{x\}, \quad x[n] \xleftrightarrow{\mathcal{UZ}} \mathcal{X}(z)$$

Note $\mathcal{UZ}\{x\} = \mathcal{Z}\{xu\}$, where u is the unit step function.

- ROC of $\mathcal{X}$ is **always** the **exterior** of a circle, including ∞
  - For rational $\mathcal{X}(z) = \frac{N(z)}{D(z)}$, we always have $\deg N \leq \deg D$
- $\mathcal{Z}\{x\} = \mathcal{UZ}\{x\}$ iff $x[n] = 0$ for $n < 0$

Examples

The calculation of unilateral z-transforms is almost the same as for bilateral z-transforms. Remember to **sum over only $n \geq 0$**.

Example. $x_1[n] = a^n u[n]$,

$$\mathcal{X}_1(z) = X_1(z) = \frac{1}{1 - az^{-1}}, \quad |z| > |a|$$

Example. $x_2[n] = a^{n+1} u[n+1] = x_1[n+1]$,

$$X_2(z) = zX_1(z) = \frac{z}{1 - az^{-1}}, \quad |z| > |a|$$

But

$$\mathcal{X}_2(z) = \sum_{n=0}^{\infty} x[n] z^{-n} = \sum_{n=0}^{\infty} a^{n+1} z^{-n} = \frac{a}{1 - az^{-1}}, \ |z| > |a|$$

Examples

The calculation of inverse unilateral z-transforms is the same as for bilateral z-transforms, but we can **only recover $x[n]$ for $n \geq 0$!**

Example. Given unilateral z-transform

$$\mathcal{X}(z) = \frac{3 - \frac{5}{6}z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{3}z^{-1})} = \frac{1}{1 - \frac{1}{4}z^{-1}} + \frac{2}{1 - \frac{1}{3}z^{-1}}$$

the **only** possibility for ROC is $|z| \geq \frac{1}{3}$. Thus

$$x[n] = \left(\frac{1}{4}\right)^n + 2\left(\frac{2}{3}\right)^n, \quad n \geq 0$$

$\mathcal{X}$ provides **no** information about $x[n]$ for $n < 0$

Example. $\dfrac{z^2}{z-a}$ is **not** a valid unilateral z-transform, since $\deg D = 1 < 2 = \deg N$

Properties of Unilateral Z-transform

Many properties are the same as for bilateral z-transform

Property	Signal	Unilateral z-transform
–	$x[n]$	$\mathcal{X}(z)$
–	$y[n]$	$\mathcal{Y}(z)$
Linearity	$ax[n] + by[n]$	$a\mathcal{X}(z) + b\mathcal{Y}(z)$
Scaling in z domain	$z_0^n x[n]$	$\mathcal{X}(z/z_0)$
Time expansion	$x_{(k)}[n]$	$\mathcal{X}(z^k)$
Conjugation	$x^*[n]$	$\mathcal{X}^*(z^*)$
Differentiation in z domain	$nx[n]$	$-z\frac{d}{dz}\mathcal{X}(z)$
Initial Value Theorem	$x[0] = \lim_{z\to\infty} \mathcal{X}(z)$	

Properties of Unilateral z-transform

Convolution Property. If $x[n] = y[n] = 0$ for $n < 0$, then

$$(x * y)[n] \xleftrightarrow{\mathcal{UZ}} \mathcal{X}(z)\mathcal{Y}(z)$$

Proof. Note $(x * y)[n] = 0$ for $n < 0$ and $\mathcal{UZ}\{x\} = \mathcal{Z}\{x\}$ if $x = xu$.

$$\mathcal{UZ}\{x * y\} = \mathcal{Z}\{x * y\} = \mathcal{Z}\{x\}\mathcal{Z}\{y\} = \mathcal{UZ}\{x\}\mathcal{UZ}\{y\}$$

Caution. The assumption $x[n] = y[n] = 0$ for $n < 0$ is important!

Example. $x[n] = \delta[n] - \delta[n-1]$, $y[n] = \delta[n+1]$,

$$\mathcal{X}(z) = 1 - z^{-1}, \quad \mathcal{Y}(z) = 0,$$

Note $(x * y)[n] = \delta[n+1] - \delta[n]$, and

$$\mathcal{UZ}\{x * y\} = -1 \neq \mathcal{X}(z)\mathcal{Y}(z)$$

Properties of Unilateral z-transform

Time delay.

$$x[n-m] \xleftrightarrow{\mathcal{UZ}} x^{-m}\left(\mathcal{X}(z) + \sum_{k=1}^{m} x[-k]z^{k}\right)$$

Proof.

$$\begin{aligned}\mathcal{UZ}\{x[n-m]\} &= \sum_{n=0}^{\infty} x[n-m]z^{-n} \\ &= z^{-m}\sum_{k=-m}^{\infty} x[k]z^{-k} \\ &= z^{-m}\left(\mathcal{X}(z) + \sum_{k=-m}^{-1} x[k]z^{-k}\right)\end{aligned}$$

Properties of Unilateral z-transform

Time advance.

$$x[n+m] \overset{\mathcal{UZ}}{\longleftrightarrow} z^m \left(\mathcal{X}(z) - \sum_{k=0}^{m-1} x[k] z^{-k} \right)$$

Proof.

$$\begin{aligned}
\mathcal{UZ}\{x[n+m]\} &= \sum_{n=0}^{\infty} x[n+m] z^{-n} \\
&= z^m \sum_{k=m}^{\infty} x[k] z^{-k} \\
&= z^m \left(\mathcal{X}(z) - \sum_{k=0}^{m-1} x[k] z^{-k} \right)
\end{aligned}$$

Linear Constant-coefficient Difference Equations

Consider difference equation

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

with initial condition $y[-1], y[-2], \ldots, y[-N]$ and causal input, i.e. $x[n] = 0$ for $n < 0$.

Take unilateral z-transform of both sides

$$\sum_{k=0}^{N} a_k z^{-k} \left[\mathcal{Y}(z) + \sum_{\ell=-k}^{-1} y[\ell] z^{-\ell} \right] = \sum_{k=0}^{M} b_k z^{-k} \mathcal{X}(z)$$

so

$$\mathcal{Y}(z) = \underbrace{\frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}} \mathcal{X}(z)}_{\text{zero-state response}} \underbrace{- \frac{\sum_{k=0}^{N} a_k z^{-k} \sum_{\ell=-k}^{-1} y[\ell] z^{-\ell}}{\sum_{k=0}^{N} a_k z^{-k}}}_{\text{zero-input response}}$$

Example

Consider difference equation

$$y[n] + 2y[n-1] = x[n]$$

with initial condition $y[-1] = \beta$ and $x[n] = \alpha u[n]$.

Take unilateral z-transform of both sides

$$\mathcal{Y}(z) + 2z^{-1}(\mathcal{Y}(z) + y[-1]z) = \mathcal{X}(z) = \frac{\alpha}{1-z^{-1}}$$

so

$$\mathcal{Y}(z) = \underbrace{\frac{1}{1+2z^{-1}} \cdot \frac{\alpha}{1-z^{-1}}}_{\text{zero-state response}} + \underbrace{\frac{-2\beta}{1+2z^{-1}}}_{\text{zero-input response}}$$

Example (cont'd)

$$\mathcal{Y}(z) = \underbrace{\frac{1}{1+2z^{-1}} \cdot \frac{\alpha}{1-z^{-1}}}_{\text{zero-state response}} + \underbrace{\frac{-2\beta}{1+2z^{-1}}}_{\text{zero-input response}}$$

Zero-input response. $\alpha = 0$

$$y_{zi}[n] = -2\beta(-2)^n, \quad n \geq 0$$

Zero-state response. $\beta = 0$, system at initial rest

$$\mathcal{Y}_{zs}(z) = \frac{1}{1+2z^{-1}} \cdot \frac{\alpha}{1-z^{-1}} = \frac{2\alpha/3}{1+2z^{-1}} + \frac{\alpha/3}{1-z^{-1}}$$

$$y_{zs}[n] = \frac{2\alpha}{3}(-2)^n + \frac{\alpha}{3}, \quad n \geq 0$$

Example

Consider difference equation

$$y[n] - y[n-1] - 2y[n-2] = x[n] + 2x[n-2]$$

with initial condition $y[-1] = 2, y[-2] = -\frac{1}{2}$ and $x[n] = u[n]$.

Take unilateral z-transform of both sides

$$\mathcal{Y}(z) - z^{-1}(\mathcal{Y}(z) + y[-1]z) - 2z^{-2}(\mathcal{Y}(z) + y[-1]z + y[-2]z^2) = (1 + 2z^{-2})\mathcal{X}(z)$$

so

$$\mathcal{Y}(z) = \underbrace{\frac{1 + 2z^{-2}}{1 - z^{-1} - 2z^{-2}} \cdot \frac{1}{1 - z^{-1}}}_{\text{zero-state response}} + \underbrace{\frac{2z^{-1}y[-1] + y[-1] + 2y[-2]}{1 - z^{-1} - 2z^{-2}}}_{\text{zero-input response}}$$

$$= \underbrace{\frac{1 + 2z^{-2}}{(1 + z^{-1})(1 - 2z^{-1})(1 - z^{-1})}}_{\text{zero-state response}} + \underbrace{\frac{4z^{-1} + 1}{(1 + z^{-1})(1 - 2z^{-1})}}_{\text{zero-input response}}$$

Example (cont'd)

For the zero-input response,

$$\mathcal{Y}_{zi}(z) = \frac{4z^{-1}+1}{(1+z^{-1})(1-2z^{-1})} = \frac{-1}{1+z^{-1}} + \frac{2}{1-2z^{-1}}$$

so

$$y_{zi}[n] = 2^{n+1} + (-1)^{n+1}, \quad n \geq 0$$

For the zero-state response,

$$\begin{aligned}\mathcal{Y}_{zs}(z) &= \frac{1+2z^{-2}}{(1+z^{-1})(1-2z^{-1})(1-z^{-1})} \\ &= \frac{1/2}{1+z^{-1}} + \frac{2}{1-2z^{-1}} + \frac{-3/2}{1-z^{-1}}\end{aligned}$$

so

$$y_{zs}[n] = \frac{1}{2}(-1)^n + 2^{n+1} - \frac{3}{2}, \quad n \geq 0$$